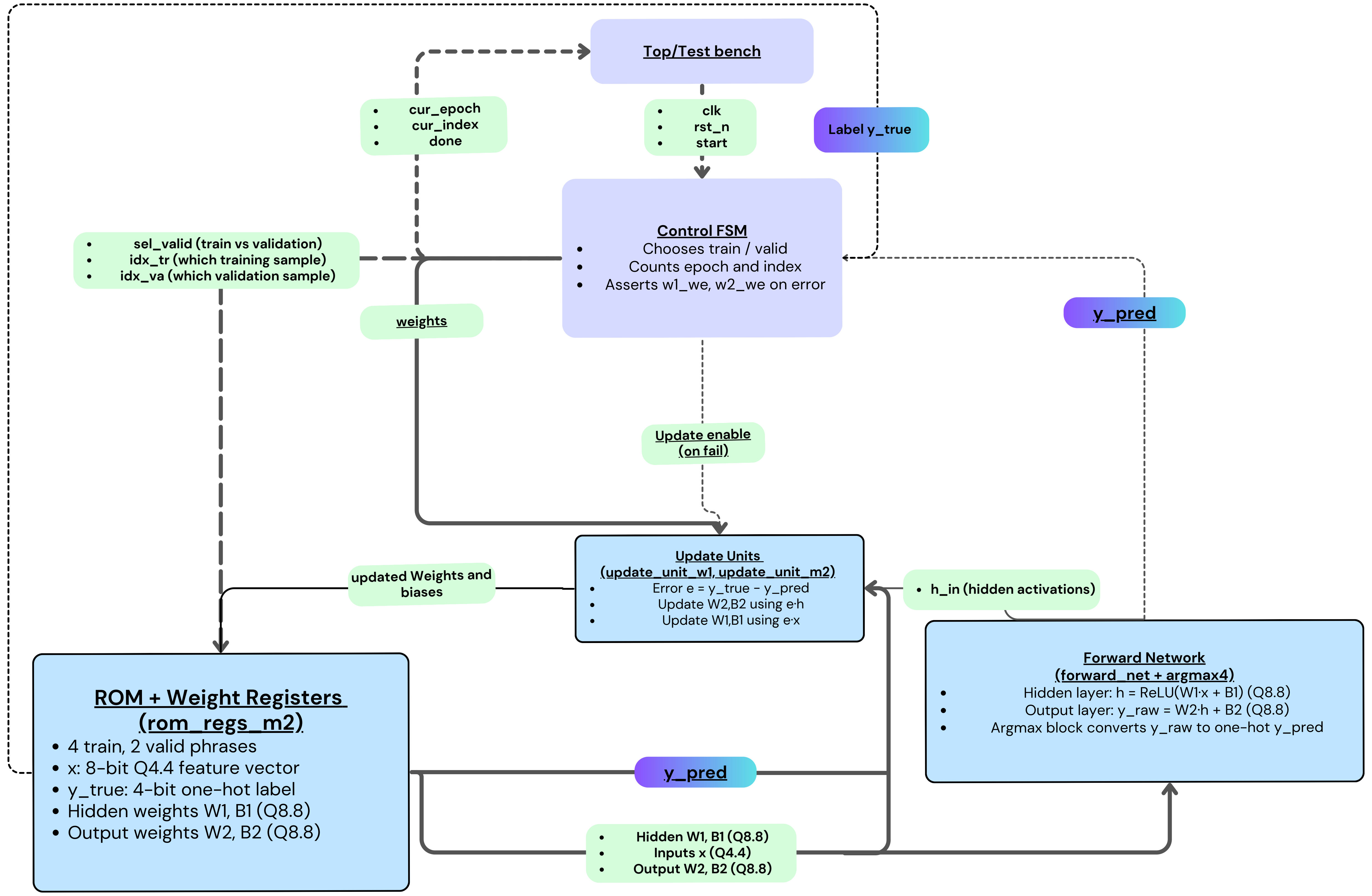
AI Project Milestone 2

By: Ahmed Elag & Marc Niven

Architecture



Control FSM
- Chooses train / valid
- Counts epoch and index
- Asserts w1_we, w2_we on error

Update Units (update_unit_w1, update_unit_m2)
- Error e = y_true - y_pred
- Update W2,B2 using e·h
- Update W1,B1 using e·x

ROM + Weight Registers (rom_regs_m2)
- 4 train, 2 valid phrases
- x: 8-bit Q4.4 feature vector
- y_true: 4-bit one-hot label
- Hidden weights W1, B1 (Q8.8)
- Output weights W2, B2 (Q8.8)

Forward Network (forward_net + argmax4)
- Hidden layer: h = ReLU(W1·x + B1) (Q8.8)
- Output layer: y_raw = W2·h + B2 (Q8.8)
- Argmax block converts y_raw to one-hot y_pred

Update Rule

ERROR
- e = y_true - y_pred ∈ {-1, 0, +1}
- Computed from one-hot labels

OUTPUT LAYER (W2, B2)
- If e ≠ 0: move weights toward the correct class
- Update uses hidden activations h
- Effective learning rate ≈ 1/4

HIDDEN LAYER (W1, B1)
- Same error sign e reused
- Update uses input features x
- Smaller learning rate ≈ 1/8

Formats

X:
-Q4.4

WEIGHTS(W1,W2):
-Q8.8

BIAS(B1,B2):
-Q8.8

SUM(FINAL OUTPUTS):
-Q8.8

MAC: HIDDEN LAYER
- w(Q8.8) * x(Q4.4) ->Q12.12
- Accumulate in Q12.12
- Output shift >>4 -> Q8.8

MAC: OUTPUT LAYER
- W2 (Q8.8) × h (Q8.8) → Q16.16
- Accumulate in Q16.16
- Shift >> 8 → y_raw (Q8.8)

EDA PLAYGROUND RUN STEPS

SETUP:

1) Site: edaplayground.com
2) Language: VHDL
3) Tools & Sim: Aldec Riveira Pro
4) Check: "Open EPWave after run"

TOP MOD & GENERICS:
- Top entity to run: testbench
- DUT inside: mini_llm_core
- Generics used: Generics used:
  - N_TRAIN = 4 (training phrases)
  - N_VALID = 2 (validation phrases)
  - MAX_EPOC = 50 (training stops early once all samples correct)

FILES:
- fx_types_m2.vhd
- relu_unit.vhd
- argmax4.vhd
- forward_net.vhd
- update_unit_m2.vhd
- update_unit_w1.vhd
- rom_regs_m2.vhd
- control_fsm_m2.vhd
- mini_llm_core.vhd
- testbench.vhd

Data

Data set

Prediction task: choose a reply for a short text input. Each sample has:
- Input sentence (simple English phrase)
- x[0..7]: 8-element Q4.4 vector (shown as 0/1 one-hot in the table)
- y_true: 4-bit one-hot label for the reply class
- Reply text:
  - • 0001 → "Hey there!"
  - • 0010 → "Take a short break."
  - • 0100 → "Get some rest."
  - • 1000 → "Good night!"

PHASE	INDEX	INPUT SENTENCE	8-BIT INPUT ENCODING	TARGET REPLY	REPLY ENCODING
TRAIN	0	HELLO	10000000	HI THERE	0001
TRAIN	1	I'M STRESSED	01000000	TAKE A SHORT BREAK	0010
TRAIN	2	I'M TIRED	00100000	GET SOME REST	0100
TRAIN	3	I'M GOING TO SLEEP	00010000	GOOD NIGHT	1000
VALIDATE	0	HELLO + IM STRESSED	11000000	TAKE A SHORT BREAK	0001
VALIDATE	1	IM TIRED + I WILL GO SLEEP	00110000	GOODNIGHT	0100

Training Data set

Chose the following 4 for training:

PHASE	INDEX	INPUT SENTENCE	8-BIT INPUT ENCODING	TARGET REPLY	REPLY ENCODING
TRAIN	0	HELLO	10000000	HI THERE	0001
TRAIN	1	I'M STRESSED	01000000	TAKE A SHORT BREAK	0010
TRAIN	2	I'M TIRED	00100000	GET SOME REST	0100
TRAIN	3	I'M GOING TO SLEEP	00010000	GOOD NIGHT	1000

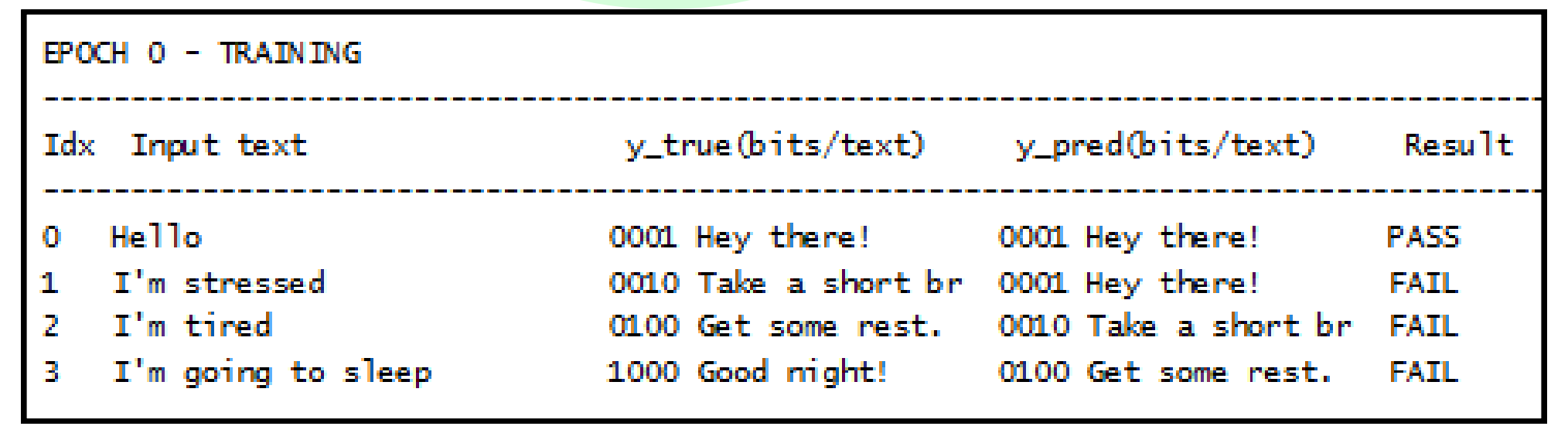
Validation Data set

Chose the following 2 for validation:

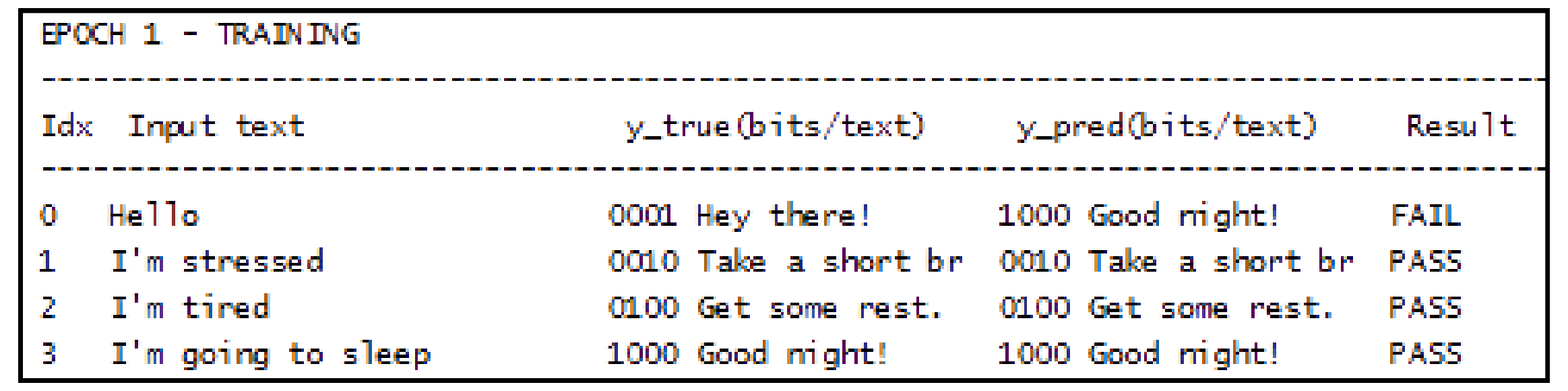
PHASE	INDEX	INPUT SENTENCE	8-BIT INPUT ENCODING	TARGET REPLY	REPLY ENCODING
VALIDATE	0	HELLO + IM STRESSED	11000000	TAKE A SHORT BREAK	0001
VALIDATE	1	IM TIRED + I WILL GO SLEEP	00110000	GOODNIGHT	0100

Output

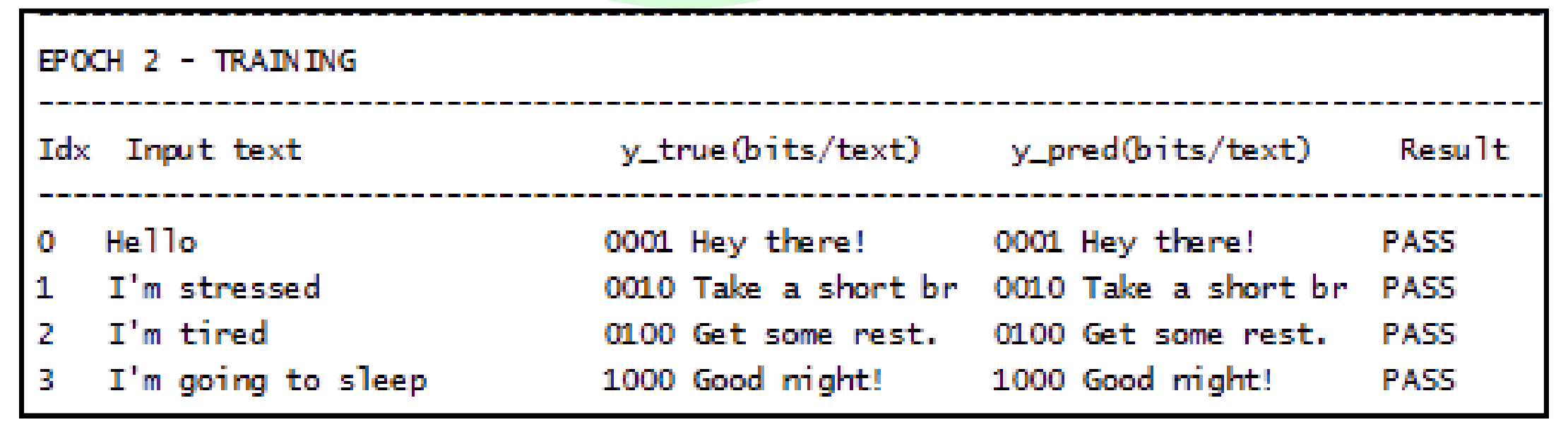
START → EPOCH = 0 → EPOCH = 1 → EPOCH = 2 → validation



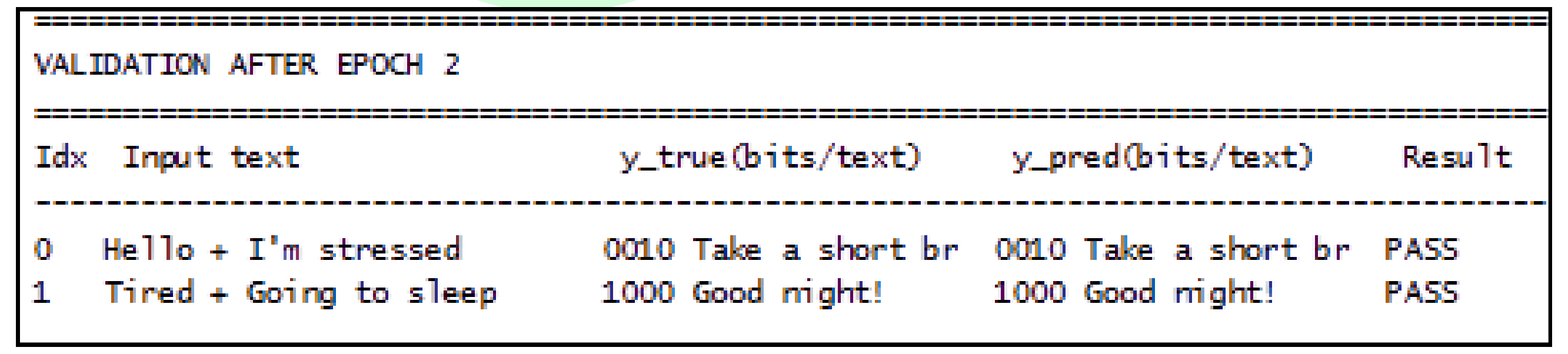
```
EPOCH 0 - TRAINING
------------------------------------------------------------------------
Idx  Input text               y_true(bits/text)   y_pred(bits/text)   Result
------------------------------------------------------------------------
0    Hello                    0001 Hey there!     0001 Hey there!     PASS
1    I'm stressed             0010 Take a short br 0001 Hey there!    FAIL
2    I'm tired                0100 Get some rest. 0010 Take a short br FAIL
3    I'm going to sleep       1000 Good night!    0100 Get some rest. FAIL
```



```
EPOCH 1 - TRAINING
------------------------------------------------------------------------
Idx  Input text               y_true(bits/text)   y_pred(bits/text)   Result
------------------------------------------------------------------------
0    Hello                    0001 Hey there!     1000 Good night!    FAIL
1    I'm stressed             0010 Take a short br 0010 Take a short br PASS
2    I'm tired                0100 Get some rest. 0100 Get some rest. PASS
3    I'm going to sleep       1000 Good night!    1000 Good night!    PASS
```

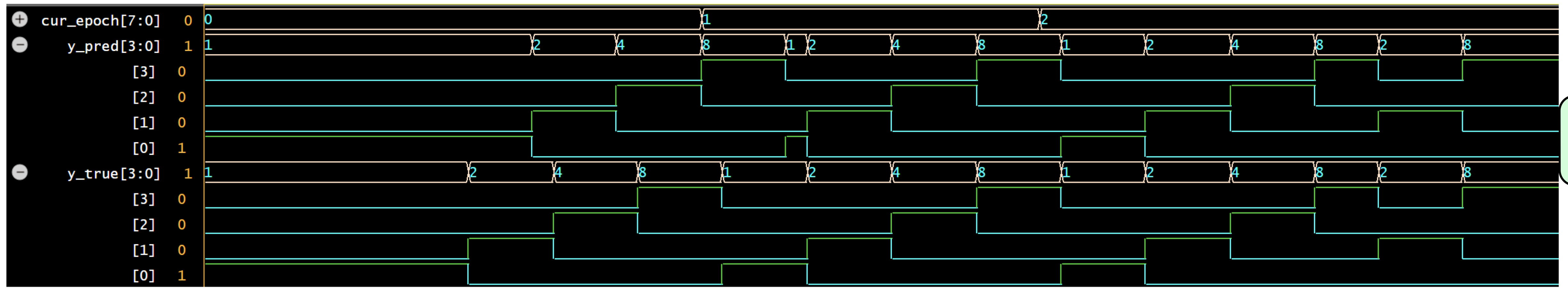


```
EPOCH 2 - TRAINING
------------------------------------------------------------------------
Idx  Input text               y_true(bits/text)   y_pred(bits/text)   Result
------------------------------------------------------------------------
0    Hello                    0001 Hey there!     0001 Hey there!     PASS
1    I'm stressed             0010 Take a short br 0010 Take a short br PASS
2    I'm tired                0100 Get some rest. 0100 Get some rest. PASS
3    I'm going to sleep       1000 Good night!    1000 Good night!    PASS
```

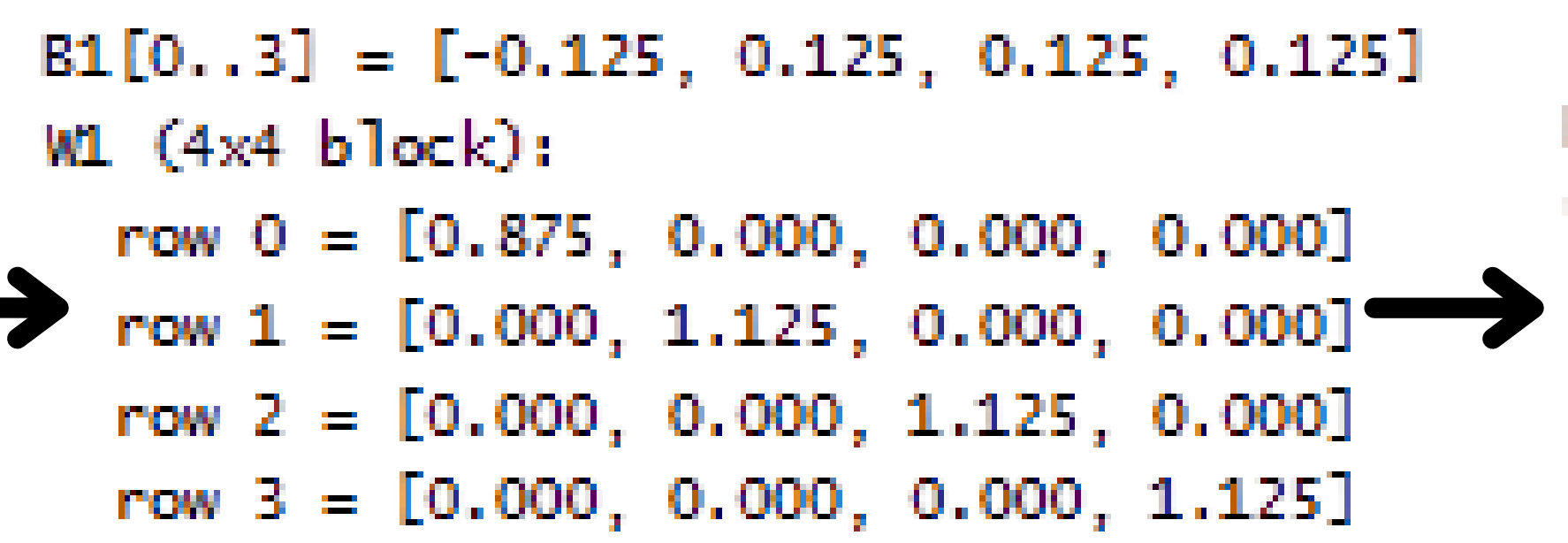


```
========================================================================
VALIDATION AFTER EPOCH 2
========================================================================
Idx  Input text                 y_true(bits/text)    y_pred(bits/text)    Result
------------------------------------------------------------------------
0    Hello + I'm stressed       0010 Take a short br 0010 Take a short br PASS
1    Tired + Going to sleep     1000 Good night!     1000 Good night!     PASS
```

Output waves: →

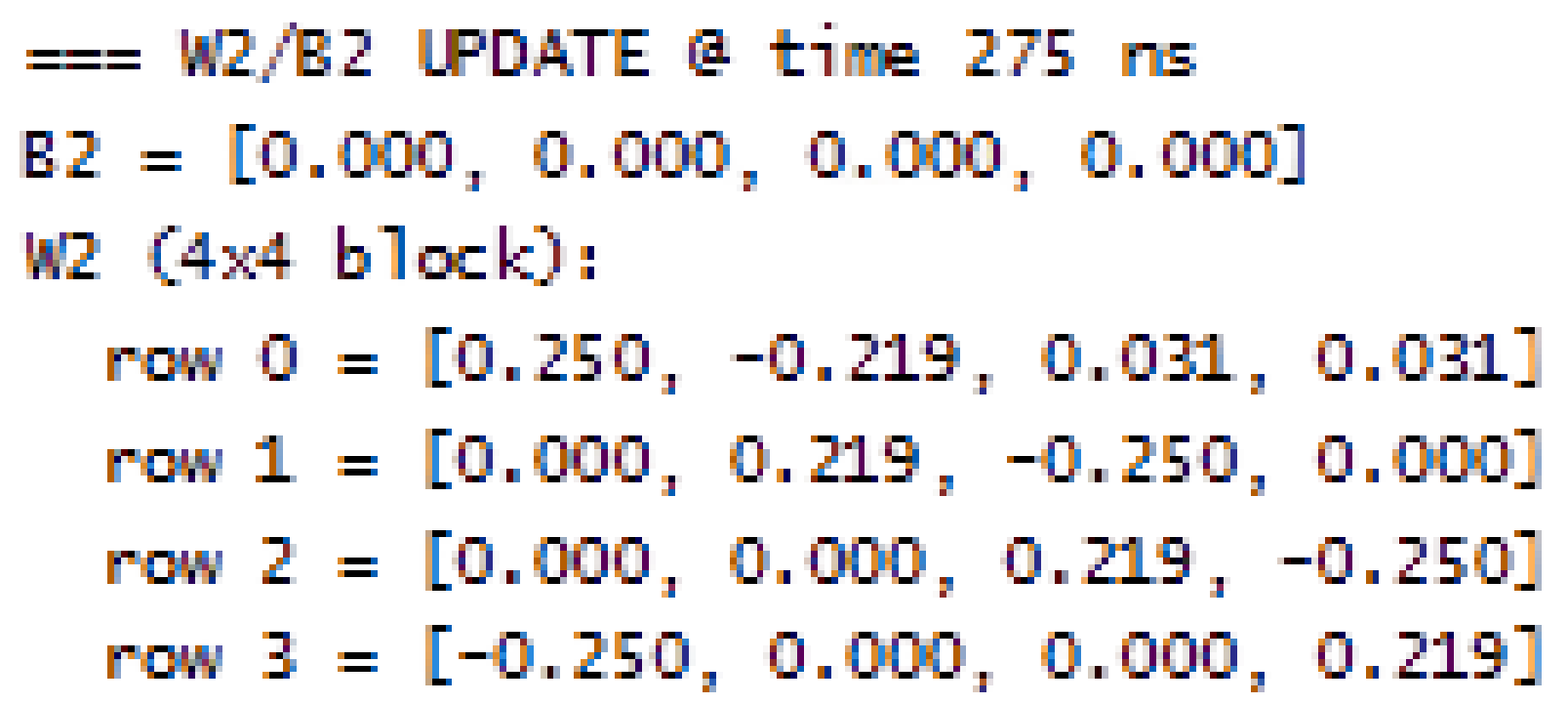


FINAL VALUES: →



```
B1[0..3] = [-0.125, 0.125, 0.125, 0.125]
W1 (4x4 block):
  row 0 = [0.875, 0.000, 0.000, 0.000]
  row 1 = [0.000, 1.125, 0.000, 0.000]
  row 2 = [0.000, 0.000, 1.125, 0.000]
  row 3 = [0.000, 0.000, 0.000, 1.125]
```

→



```
=== W2/B2 UPDATE @ time 275 ns
B2 = [0.000, 0.000, 0.000, 0.000]
W2 (4x4 block):
  row 0 = [0.250, -0.219, 0.031, 0.031]
  row 1 = [0.000, 0.219, -0.250, 0.000]
  row 2 = [0.000, 0.000, 0.219, -0.250]
  row 3 = [-0.250, 0.000, 0.000, 0.219]
```

→

STOP AT EPOCH = 2

The y_pred and y_true converge when epoch = 2

Therefore, no need to continue and validation complete

CITATION

OpenAI. (2025, October 27). ChatGPT (GPT-5) [Large language model]. https://chat.openai.com/